CAESAREAN When normal vaginal delivery is not possible due to factors like position of the baby and nature of the placenta, the baby is delivered through a surgical incision in the woman's abdomen and uterus. It is also termed as abdominal delivery or Caesarean Section or 'C' Section.

through the vagina, followed by the foetus. The placenta along with the remains of the umbilical cord called **"after birth"** is expelled out after delivery.

Lactation is the production of milk by mammary glands. The mammary glands show changes during every menstrual cycle, during pregnancy and lactation. Increased level of oestrogens, progesterone and **human Placental Lactogen (hPL)** towards the end of pregnancy stimulate the hypothalamus towards prolactin – releasing factors. The anterior pituitary responds by secreting **prolactin** which plays a major role in lactogenesis.

Oxytocin causes the "**Let-Down" reflex**- the actual ejection of milk from the alveoli of the mammary glands. During lactation, oxytocin also stimulates the recently emptied uterus to contract, helping it to return to pre - pregnancy size.

Colostrum

Colostrum, a nutrient rich fluid produced by the human female immediately after giving birth, is loaded with immune, growth and tissue repair factors. It acts as a natural antimicrobial agent to actively stimulate the maturation of the infant's immune system. No artificial feed can substitute the first milk, with all its natural benefits and therefore should be definitely fed to the baby after birth.

The mammary glands secrete a yellowish fluid called **colostrum** during the initial few days after parturition. It has less lactose than milk and almost no fat, but it contains more proteins, vitamin A and minerals. Colostrum is also rich in **IgA** antibodies. This helps to protect the infant's digestive tract against bacterial infection. Breast milk is the ideal food for infants as it contains all the constituents in suitable concentration and is easily digestible. It is fully sufficient till about 6 months of age and all infants must be breast fed by the mother to ensure the growth of a healthy baby.

Summary

Reproduction is a process which helps in the continuity and maintenance of a species. Human beings are sexually reproducing and viviparous. The reproductive events include gametogenesis, insemination, Fertilization, cleavage, implantation, placentation, gastrulation, organogenesis and parturition.

The female reproductive system consists of a pair of ovaries, a pair of oviducts, uterus, cervix, vagina and external genitalia. The male reproductive system consists of a pair of testes, a pair of duct system, accessory glands and external genitalia called penis.

The process of formation of gametes in the male is called spermatogenesis and in the female is called oogenesis. The reproductive cycle in females is called menstrual cycle and it is initiated at puberty. The ovum released during the menstrual cycle is fertilized by the sperm and the zygote is formed.

Zygote undergoes repeated mitotic division and the blastocyst is implanted on the walls of the uterus. It takes about 280 days or 40 weeks for the entire development of the human foetus and it is delivered out through the process of child birth or parturition. The new born baby is breast fed by the mother.

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